Ibasho agent blackboard

State (orchestrator, kept current)

orchestrator state · active
Phase: M-A first light plus independent M-B work, light testing. Full history: `docs/archive/BLACKBOARD-to-M0.typ`.
Done: G0 scaffold, M-0 slice (physics, rules, search and money, Lean spec and checker with TS mirror, narrator and memory, scene, agent link, first app screen, content and landing, hero assets), SPEC 14 contracts in core.
In progress: WP-01 physics (A1 to A10, maintenance decay), WP-06 UI (M-A and M-B components), WP-07 scene (review), WP-08 app (shell, onboarding, Reveal with scene), WP-03 rules (M-B), WP-02 money and score (M-B, SPEC 14 run and negatives), WP-04 checker gaps (bills, score, memory.completed).
Idle until M-D: WP-05 AI, WP-09 content, WP-12 agent link, WP-10 QA (testing is light for now), WP-11 assets.

Contract index (where the source of truth lives)

- Contracts and zod schemas: `@ibasho/core` (`packages/core/src/**`, `@ibasho/core/schemas`). SPEC 14 kinds: `maintenance`, "If nothing is done" run, per-package `negatives`.
- Measure and option ids: `packs/home-uk/src/ids.ts` (frozen; GH1 432 heavy, 10,800 packages).
- Entry points: packs export `./engine`, `./ui`, `./ids`; apps start at `src/main.tsx`; `packages/fixtures` exports `./*` (no index).
- Certificate, constants, seals, spec hash: generated from Lean into `@ibasho/spec-gen` (real, not a placeholder).
- Ownership as code: `tools/lint/ownership.ts`. `ACKNOWLEDGEMENTS.md`, the blackboard, STATUS and `docs/refs.yml` are shared append.
- Agent link: `node apps/mcp/src/main.ts` (no build); `pnpm agent:install`.

Decisions (condensed; details in the archive)

orchestrator decision · final

- Stack: pnpm monorepo, Vite, React, TypeScript 6.0.3, R3F, IndexedDB, Biome, Lean 4 v4.34.0 core only. Local first, offline first. Versions in ARCH 13.
- Plan of record: VISION, SPEC (rulings R1 to R24), ARCHITECTURE, DESIGN_SYSTEM (Andon), BUILD_PLAN; red-team rulings in `docs/review/DECISIONS.md` override older text.
- Hero home GH1: 1905 London mid-terrace in a conservation area. Score weights F 1/5, C 1/5, E 3/20, A 3/20, H 1/10, R 1/10, T 1/10, from Standard mode; shown as a number with four groups, no letter. Futures: Least upfront, Best return, Lowest running carbon.
- Lean owns formulas and predicates; pack data travels in certificates; the checker is proved sound and the TS mirror matches it.
- Engine determinism: no inexact `Math` in engine code, `dmath` instead, fixed point after quantise.
- Narrator is a plugin: template always on, Ollama local (`qwen3:8b` default; `:cloud` models count as cloud and go in the outbound ledger), Claude later.
- Agent link (WP-12): MCP server, bridge mounted by the companion, Claude Code plugin, installer. Companion: loopback host, Origin allow list, Bearer token for scripts.
- Bad outcomes (SPEC 14): "If nothing is done" line, custom what-if requests as typed ops, negatives beside savings.
- Reuse proven open source and credit it in `ACKNOWLEDGEMENTS.md`. Asset licensing is not a constraint.
- Testing is light for now: each WP runs only its own package tests. No gates until the owner asks.

orchestrator decision · final
Owner direction (2026-10-11): prototype first, outside in. A high-fidelity, responsive prototype of every user flow (new, revisit, restart, current plus what-if, update) with an honest staged progress bar for rendering and computing, on fixture data shaped like real engine output and labelled Preview. Then replace fixtures with real computation stage by stage. See `docs/research/v3/00-owner-direction.md`.

director decision · final
v3 plan of record (2026-10-11): `docs/PLAN-V3.md`, with SPEC 15 to 26 (rulings R25 to R45), ARCH 15, BUILD_PLAN 9 to 12 and DESIGN_SYSTEM 13. A studio for eight building types in a real 3D city, Hyderabad

first (hero: Vindhya Bhavan, IIIT). Order: V-0 contracts (WP-00, WP-10), then wave 1 V-P prototype on Preview data (design WP-06; developers WP-08, WP-15, WP-07, WP-13, WP-14, WP-17; QA WP-22 product, WP-23 UX, WP-24 content, review based), then real stages V-1 to V-6 in progress-bar order, then V-7 partner and V-H hero. New WPs WP-13 to WP-24 own disjoint paths (BUILD_PLAN 10). Testing stays light.

Claims

steward:WP-00 claim · done
SPEC 14 contracts in `packages/core/**`: the `maintenance` plugin kind, the "If nothing is done" run, per-package negatives, and the deferred WP-02 `OwnerCtx` fields.

builder:WP-08 claim · active
Gate M-A (app shell and store): `apps/web/**` except `apps/web/public/assets/**`, `packages/store/**`. Shell layout zones (left rail, top bar with home switcher and lantern, mobile tab bar), `/app/start` onboarding in 5 typed steps with no budget question, Reveal on the home stage with WP-07 `Stage` (SVG fallback without WebGL), packs imported by package name.

builder:WP-06 claim · active
Gates M-A and M-B UI kit: `packages/ui/**`. M-A: `Sheet`, `NarratorStrip`, `ScoreGroups`, `EpcChip`. M-B: `Futures` (with the fourth line "If nothing is done" and a negatives list on every card), `MeasureCard`, `FuelBars`, `Slider`, `WhyNot`, `CheckLane`, `MonthRing`, plus the chart frame with `TableAlt`, a first mon set and `GradeBadge`. Stories for each.

builder:WP-02 claim · active
M-B pieces without new physics: score with exact Q and the four groups, EPC band proxy, IRR, paybacks, loan stub, flips, the "If nothing is done" run and per-package negatives (SPEC 14), costs for every SPEC 9 option. Paths: `packages/engine/src/{api,index}.ts`, `packages/engine/src/{search,finance,score,flips,evidence}/**`, `packages/engine/{test,scripts}/**`, `packs/money-uk/**`, `packages/fixtures/src/results/**`.

builder:WP-03 claim · done
M-B (C1 to C7): `packages/engine/src/rules/**`, `packs/rules-uk/**`, `packs/rules-in/**`. Named functions, `noise()`, the SPEC 9 `rules-uk` set with `APPL.*`, sentences, obligations, rule lint.

agent-A:WP-00 claim · active
v3 platform and data (Agent A). WP-00 V-0 contracts (`packages/core/src/v3/**`, schemas, layer rules, installs), WP-10 `tools/lint/ownership.ts` v3 map, WP-13 `packages/preview` and `packages/fixtures/src/preview`, WP-14 `packages/plan` and `packages/fixtures/src/buildings`, WP-17 `packs/city-hyderabad`, `packs/city-london`, `tools/city`. Not touching B (ui, web, store, editor) or C (scene, packs ui halves).

builder:WP-06+WP-08+WP-15 (agent B) claim · done
V-P prototype, product UI. `packages/ui/**` (v3 kit: StageBar, PreviewStamp, ScaleLadder, LensPanel and AnalysisLegend, eight type cards and mons, EuiGauge, EndUseBar, RatingLadder, LedgerRow, CompareGrid, FilmStrip, ModeTabs), `apps/web/**` except `public/assets` (studio shell, modes Make Look Choose Prove Tell, the five flows on Preview data), `packages/store/**` (Dexie v2 with migration), `packages/editor/**` (new, WP-15: plan editor). Builds against `@ibasho/core` v3 contracts in the working tree; Preview results come from a local adapter in `apps/web/src/studio/preview/` until `packages/preview` lands, then switch is one import.

builder:WP-07 (agent C) claim · done
V-P city scene: `packages/scene/src/city/**` (new `CityStage`, stand in campus context and Vindhya stand in, sun and time, lenses, penumbra, phototropic variant, lanterns, SVG fallback), `packages/scene/dev/**`, `packages/scene/test/**`. Reads `@ibasho/core` v3 contracts only; does not edit core, fixtures or city packs.

Interfaces (live only)

builder:WP-02 interface · open
Requests (non-blocking): (1) WP-03: `BUILTIN.MEMORY` lives in the search (`packages/engine/src/search/run.ts`)

because the evaluator has no memory facts; move it into `BUILTIN_RULES` when a fact form exists, keeping the id. The budget rule receives the gross central cost as `p.capex`; say if Lean `Budget.lean` should compare N0 instead. (2) WP-01: heat pump costs size the unit with a cost-side estimate (`heatPumpSizeKw` in money-uk); expose `hp.size_kw` to cost plugins at M-A, e.g. a `size` argument or a twin field. (3) WP-00: `OwnerCtx` needs fuel escalation scenarios and a confirmed counterfactual replacement (A_cf) by M-B; M-0 uses 0% escalation and no A_cf.

builder:WP-09 interface · open

Chapter `requires` convention: a requirement is a plugin id or `kind:<PluginKind>` (any plugin of that kind, checked with `registry.all(kind)`); `presenceOf(registry)` adapts a `PluginRegistry`. Current landing requires: drawn `climate-london` + `kind:archetype`; what-if `loft`, `ashp`, `kind:tariff`; futures `kind:search-strategy`, `rules-uk`; proof `kind:checker-rung`. **Request to WP-08:** the landing discovers plugins at build time from `packs/*/src/{engine,ui}/module.ts` (`apps/landing/src/discover.ts`), because `apps/web/src/manifest.ts` is still empty. Once the manifest lists the registered modules, please keep it importable from Node (plain `.ts`, no JSX) or export the module list as data, and WP-09 will switch to it so the page never advertises a pack the app does not load. **Links:** "Step into your home" goes to `/app/start`; "See a sample home" goes to `/app/home/gh1` (GH1 read-only, Story mode). WP-08: tell me if the sample route differs.

builder:WP-09 interface · open

To WP-10 (smoke): `apps/landing` now prerenders the real page: `tests/smoke/boot.spec.ts` will fail on `main[data-andon="blank"]` being empty. Please assert `main[data-andon="landing"]` visible, `section[data-chapter="hero"]` present, title `Ibasho`, body background still `rgb(244, 238, 227)` in light scheme, no console errors. **To WP-06 (fonts):** `@ibasho/ui/fonts.css` compiles to 848 `@font-face` rules (714 kB CSS); the landing loads it after first paint to keep AC-L1, but a Latin-first subset entry (for example `./fonts-latin.css`) would help both apps. Small woff slices also get inlined as data URIs by Vite's default `assetsInlineLimit`; the landing turns that off for fonts.

builder:WP-12 interface · open

Request to WP-05 (companion mount, one line plus discovery): add `@ibasho/agent-bridge` to `apps/companion` dependencies at the next install window, then `app.route(BRIDGE_MOUNT, createBridgeApp({ hub, token }))` with `token = createServerToken()` and `hub = new AgentHub({ onLedger })`; bind `BRIDGE_HOST` (127.0.0.1); on listen `writeServerJson({ port, token, pid: process.pid, startedAt, bridge: BRIDGE_MOUNT })` and print `hub.pairingCode().code`; `removeServerJson()` on clean exit. All exported from `@ibasho/agent-bridge/server` (constants from `@ibasho/agent-bridge`); snippet in `packages/agent-bridge/README.md`. The SSE transport needs no extra dependency; `/link/ws` turns on if you pass `upgradeWebSocket` from `@hono/node-ws`.

builder:WP-12 interface · open

Request to WP-08 (Settings > Agent link): add `@ibasho/agent-bridge` to `apps/web` at the next install window. Toggle off by default with one line saying a connected agent may send what it reads to its model provider (SPEC 13); a code field; then `connectAgentLink({ origin: 'http://127.0.0.1:7357', pair: { code }, client: { app: 'ibasho-web', version }, handlers, onStatus, onLedger })`. `handlers` implements `LiveHandlers` (`packages/agent-bridge/src/protocol.ts`): `apply` runs ops through the OpValidator and the undo stack with an "agent" mark; `say` runs the narrator validator; `screenshot` returns a base64 PNG of the stage; `thread` draws the red thread. Call `link.pushState(state)` on route, chapter, layer and selection changes; keep `link.session` in sessionStorage to reconnect after a reload (`pair: { session }`); write every `onLedger` entry to the outbound ledger (SPEC 5.14). Every incoming call is already schema-checked in the client.

builder:WP-04 interface · proposed

To WP-02 (engine emitter), from WP-04: the certificate now has optional slots the checkers need (`@ibasho/spec-gen cert.ts`): `groups[].optionMeta` (`{ [optionId]: { measure, kind } }`), `context.home` (`{ region, facts: homeFacts(twin), checks: twin.checks }`), `claimed.saving` (int64 [row][fuel] annual Wh, before minus after) and `claimed.carbonAnnual` (int64 [row] g, `carbonSavingG`). Until the engine emits them, `@ibasho/verify` `emit(result, { optionMeta, home })` fills them (saving and carbon from the certificate physics). Please emit them from `emitCertificate` so the checks test the engine's own claims. Also: `SPEC_HASH` is now real (`ab8d651f...`), so the result hash changed; run `pnpm --filter @ibasho/engine gen:results` to refresh `packages/fixtures/src/results/gh1-m0.search.json` (engine `results.test.ts` and mcp `l2-engine.test.ts` are red until then).

reviewer:WP-04 interface · open

To WP-02 and WP-04 (M-P): `context.memory.completed`. SPEC AC-P1 says a measure marked done leaves future searches. Today the engine copies `memory.completed` into the certificate but does not read it, and neither `ibasho-check` nor the mirror checks it, so a certificate offering a completed measure passes. Proposal: once WP-02 defines the semantics (exclude like `rejected`, or fold into the twin), WP-04 adds it to `deriveRow` in Lean and the mirror, with a mutant.

builder:WP-04 interface · proposed

To WP-02 (engine emitter), checker gaps closed: the checkers now test three more claims. (1) `money.bills`: `claimed.bills`, int64 [row][fuel] annual your-home bill in micro-pence (slab-exact, `billOf` of the row's delivered energy, standing charge rule as in your certificate entry). Missing means the check fails, so `@ibasho/verify` `emit` fills it from the certificate physics until `emitCertificate` does. (2) `score.recompute`: optional `claimed.score = { zone: 'uk' | 'in', mode: 'burden' | 'per-area', rows: [{ row, indicators, sub, groups, total }] }`, rationals as `Q.toWire` strings, rows strictly increasing, and every front and pick row present; sub, groups and total must equal `subScore`, `groupScore`, `score` of the indicators exactly. Without it the check is skipped with its reason. `emit(result, { ..., score: { zone, mode, indicators } })` and `scoreClaim(...)` build the block from indicators until the engine emits its own. (3) `context.memory.completed` (SPEC AC-P1): an option id, or an option of a measure id, in `completed` is excluded exactly like `rejected` (status blocked, `BUILTIN.MEMORY`). The engine copies `completed` but does not exclude it yet, so a real search with a done measure now fails `feas.recompute`; please treat `completed` like `rejected` in search. This answers the reviewer entry on `memory.completed`.

steward:WP-00 interface · final

Bad outcomes contracts (SPEC 14), additive, in `@ibasho/core` (`packages/core/src/maintenance.ts`):

- **`maintenance` plugin kind** (engine, numeric, so provenance with sources and grade is required). `MaintenancePlugin`: `ages` (`{ kind: 'measure', measureId }` or `{ kind: 'system', category, systemIds? }`), `neglect` (message key), `effects: DecayEffect[]`, optional `endOfLife: { ageYears: { lo, mid, hi }, replaceWithSystemId?, explain }`. `DecayEffect`: `param` (JSON Pointer into the twin, as in `TwinPatch`), `unit`, `rate` (`{ form: 'relative', ppbPerYear: { lo, mid, hi } }` with bigint ppb, same recurrence as discounting, negative means the value falls; or `{ form: 'absolute', perYear: { lo, mid, hi } }` in `Q`), `onsetYears?`, `onsetFrom?: 'install' | 'as-of'`, `clamp?: { min?: Q, max?: Q }`, `quantity` (the evidence quantity id of the aged value), `explain`. Registry check `maintenance` (`checkMaintenance`): ordered ranges, pointer params, a target, effects or an end of life. WP-01: decay models go in `packs/home-uk/src/engine/maintenance/`.
- **"If nothing is done" run:** optional `EngineApi.doNothing?(req: DoNothingRequest): Promise<DoNothingResult>`. Request: `twin`, `owner`, `asOf`, `mode?` (default `your-home`), `horizonYears?` (default `owner.horizonYears`), `reportYears?`, `rate?: 'lo' | 'mid' | 'hi'`. Result: `hash`, `specHash`, `maintenanceFingerprint`, `mode`, `horizonYears`, `years: { year, deliveredWh: per fuel, bill, carbon, score? }[]`, `effects: { maintenanceId, param, year, evidence, band? }[]`, `replacements: { maintenanceId, dueYear: { lo, mid, hi }, cost?, evidence }[]`, `totals: { costNpv, carbon }`, `negatives`, `notApplicable` (model ids that matched nothing), `graph`. Every value is an evidence id, never a raw number (AC-B1). WP-02 implements, WP-08 draws the fourth line, WP-12 may expose it.
- **Negatives:** `Negative { indicator, kind: 'fuel-rise' | 'money' | 'carbon' | 'performance' | 'disruption' | 'risk', evidence, before?, after?, risk?: 'moisture' | 'overheating' | 'noise' | 'fire' | 'structural' | 'other', ruleId?, maintenanceId?, message }`; `NegativesByPackage = Record<packageId, Negative[]>`. `SearchResult.negatives?` covers baseline, front and picks (like `graph`); optional `EngineApi.negatives?(resultHash, packageIds)` for the rest. AC-B3 reads these. WP-09 writes the `message` copy.
- **WP-02 request (3), now applied:** `OwnerCtx.escalation?: { scenario: 'low' | 'central' | 'high', ppbPerYear?: per fuel }` (`EscalationScenario`, `ESCALATION_SCENARIOS`; scenario values belong to the money pack) and `OwnerCtx.counterfactual?: { replacementDue: Tri, systemId?, cost? }` (A_cf counts only on `yes`, DEC SC-F5). Absent fields keep M-0 behaviour. `SystemCategory` is now a named type.
- New `EngineApi` members are optional so current implementers still compile; they become required at M-B. `PLUGIN_KINDS` has 32 entries.

builder:WP-07 interface · final

Scene entry for the app shell (WP-08). `import { HouseStage } from '@ibasho/scene'`; `<HouseStage twin={twin} quantities={q} style="washi" />` fills its parent (give it a sized box). `style`: `washi`, `yoru` or `svg` (no WebGL). Optional: `result` (quantities then come from `result.graph`), `layers` (default `heat-flow`, `zone-tint`), `pencil`

(underdrawing while computing), `reducedMotion`, `tierCap`, `llmGenerating`, `store` (shared selection), `showOutline`, `assetsBase` (default `/app/assets/`, `null` stays procedural) and `stage` (any other `Stage` prop). This replaces `apps/web/src/home/HouseDrawing.tsx` at M-A. **Pack ui (`@ibasho/pack-home-uk/ui`):** `homeUkUiModule` (default export of `src/ui/module.ts`, id `pack-home-uk/ui`) loads 16 `measure-ui` plugins, ids `home-uk.measure-ui.<measureId>`; `mon` values are keys of `MONS` in `packages/ui/src/mon/set.ts` (WP-06: please export `MONS` or `isMonId` from `@ibasho/ui` so a contract test can check them). `FoldSpec.targets` are element kinds plus `system` for plant drawn as props. `ArchetypeThumb` and `thumbnailDrawing(form)` draw every `BuiltForm` as an ink elevation (WP-08 Shape step can use them instead of local thumbs). WP-10: the ARCH 4.3 contract "every measure id has a `measure-ui`" can import `HOME_UK_MEASURE_UI`.

builder:WP-02 interface · final

M-B engine outputs (WP-08, WP-09, WP-10, WP-12): `EngineApi.flips(hash, param)` returns every crossing (`at` is p/kWh to 0.001 or percent to 0.001; a pick that empties reads `none`); `doNothing(req)` and `negatives(hash, ids)` are implemented. `SearchResult.negatives` covers the baseline, front and picks. New evidence quantities (your-home unless noted): `acct.saving.<fuel>` (annual), `money.bill_saving.<fuel>`, `fin.payback_simple` (exact years or cat `immediate`/`never`), `fin.payback_discounted`, `fin.irr` (exact percent or cat `n/a:<reason>`), `fin.loan_monthly`, `fin.cash_positive_m1`, `risk.<kind>.<ruleId>`; standard: `score.sub.<F..T>`, `score.group.<g>`, `score.total` (exact, or an interval while T is unknown), `score.delta`; rating: `epc.sap_est`, `epc.band_est` (cat `C (B to D)`). Do-nothing quantities: `<effect.quantity>.<modelId>.y<t>` (with `.lo`/`.hi`), `donothing.delivered.<fuel>.y<t>`, `donothing.bill.y<t>`, `donothing.carbon.y<t>`, `donothing.replacement_due.<id>`, `donothing.replacement_cost.<id>`, `donothing.cost_npv`, `donothing.carbon_total`. Negative message keys for WP-09: `negative.fuel-rise.<fuel>`, `negative.bill-rise.<fuel>`, `negative.year1-cost`, `negative.npv`, `negative.no-payback`, `negative.carbon-rise`, `negative.lifetime-carbon-rise`, `negative.score-fall`, `negative.band-fall`, `negative.risk.<kind>`, `negative.donothing.{fuel-rise.<fuel>,bill-rise,carbon-rise,replacement-due}`, plus each decay effect's own `explain` key. The `epc-band-at-least` target now works (central band).

builder:WP-02 interface · open

Requests from WP-02 (non-blocking): (1) WP-04: spec-gen writes integer rationals as `"4/1"`, which core `Q.fromWire` rejects; the engine parses them with `specQ`. Either emit `"4"` or say this is intended. Also export `prices/reference-v1` (SAP 10.2 Table 12 prices, standing charges, primary energy factors) so `packages/engine/src/score/reference.ts` can go. The certificate does not yet carry A_cf or escalation; both are absent unless the owner sets them, so M-0 certificates are unchanged. (2) WP-00: `EpcProxy.sap(run)` needs the floor area (money-uk adds `sapFor(run, tfaM2)`); a money-pack kind for escalation scenarios (values now SF 5.1 defaults in `finance/owner.ts`); a disruption field on `MeasureModel` (days, rooms) for disruption negatives. (3) WP-01: put PV yield and self use on `SimResult` (the R indicator uses exported energy as a lower bound); write the decay models in `packs/home-uk/src/engine/maintenance/` (the engine runs any `maintenance` plugin now; tests use a boiler model of −1% a year, end of life 12, 15, 20 years).

builder:WP-03 interface · final

Rules M-B for WP-02, WP-04, WP-08, WP-12 (all from `@ibasho/engine`, source `packages/engine/src/rules`). `evaluateRules` now uses `DEFAULT_FNS` (pass `fns: {}` for the old all-unknown behaviour), merges `K_FACTS` (`k.*`) into every env and `derivePackageFacts(options, home)` under `ctx.pkg` for package rules (`p.kind.<kind>`, `p.adf.{major,minor,now}`, `p.vent.provided`, `p.airtighter`, `p.export.{W,limitW}`, `p.envelope.share`, `p.funded.pas2035`); `packageEnv(ctx)` returns the exact package env. `homeFacts` covers RP 4 (`h.wall.u`, `h.wall.solidMasonry`, `h.art4.P14.A` style codes, `h.dhw.class`, `h.vent.ap50`, `h.floor.type`, `h.envelope.areaM2`, `h.past.adf.*`, …). `packageFacts` takes `fundedPas2035?`. Built-ins add `BUILTIN.TENURE.{TENANT,LEASE}`, `BUILTIN.PREREQ.BATTERY-PV`, `BUILTIN.CONFLICT.{EWI-IWI,GLAZING}`. Also: `normaliseObligations`, `obligationPatch(twin, obligations)` (`{ patch, unpatched }`, idempotent), `renderRule`/`renderExpr`, `lintRules`, `noise({ twin, units })`, `noiseFact(twin, units)`, `mcs020aLpLe`, `adfCategory`, `meesCapMp`, `dtHosting`, `ensRetv`.

- **WP-02:** (1) wire `EngineApi.noise` in `src/api.ts` to `noise` from `./rules/index.ts` (it returns `NoiseResult` synchronously). (2) Apply obligations with `obligationPatch` before re-running physics; map `{ kind: 'assessment', check: 'moisture-assessment' | 'pas2035-assessment' }` to the costed options `assess-moisture` and `assess-pas2035`; `emitter-flow` and `hp-meets-design-load` come back in `unpatched`. (3) Pass `fundedPas2035: true` to `packageFacts` only when a stacked grant requires PAS 2035. (4) The rule pack grew, so regenerate `gh1-m0.search.json` (statuses are unchanged; outcomes and consents grow). (5) `BUILTIN.MEMORY` and `BUILTIN.TARGET` stay in the search.

- **WP-04 and WP-02 (certificate parity):** package rules now read derived `p.*` facts, and `APPL.*`/built-in blocks depend on them. The certificate should carry `packageEnv(ctx)` (or the derived facts) per row so `feas.recompute` sees the same env. Named functions are used only in `consent`/`oblige`/`warn`/`info` rules (lint `fn-status`), so status never depends on `NAMED_FN_IMPLEMENTED`. Lean mirrors to add when convenient: `mees_cap`, `dt_hosting`, `ens_retv` (TS is exact rationals; contracts in `fns/*.ts` headers).
- **WP-01:** emit the option facts listed in `OPTION_FACTS` (`@ibasho/pack-rules-uk/engine`), above all `m.ashp.noise = noiseFact(twin, units)`, `m.ashp.{volumeM3,units,mount,frontsHighway,nearerHighway,coolingOnly}`, `m.wall.share`, `m.ewi.{finish,matches}`, `m.pv.{mount,exportW,frontsHighway,projectionM}`, `m.battery.exportW`. Until then those rules read unknown and list their consent (worst case), which is honest.
- **WP-09:** message keys used by rules are `rules.<rule id lowercased>.explain` plus each effect's `reason`/`risk`/`note` key; `renderRule` gives a fallback English sentence.

agent-A:WP-00 interface · final

V-0 contracts v3: what B and C can import now (typecheck green; all additive, every v2 type unchanged).

- **`@ibasho/core`** (barrel re-exports `packages/core/src/v3/**`): building `BuildingDoc`, `Storey`, `Room2D`, `Segment`, `WindowRule`, `ShadeRule`, `Opening`, `Room3D`, `Core`, `RoofSpec`, `FreeShade`, `ThermalZone`, `MassingSpec`, `TenureSplit`, `Mm`, `Mdeg`, `Mm2`, `PointMm`, `PolygonMm`, `BuildingTypeId`/`BUILDING_TYPES`, `FacadeBin`/`FACADE_BINS`/`facadeBin(mdeg)`; compiled `CompiledBuilding`, `ZoneTwin`, `Quantities`, `Adjacency`, `PlanCheck`, `SurfaceSet` (triangulated mm surfaces with ids for scene and site), `isZoneTwin`. Place `Place`, `ContextBuilding` (height `lo/likely/hi` mm plus `source`), `ContextTree`, `CityTile` (`ibasho.city/1`), `CityPackPlugin`, `tileKey`. Site `SiteResult`, `Epw`, `Resolution`, `RESOLUTION_LEVELS`, `DaylightTier`, `SunHours`. Results `BuildingResult`, `EndUse`/`END_USES` (normative order, PV negative), `TariffBill`, `WaterBalance`, `ComfortHours`, `CodeRow`, `RatingRow`, `BaselineRow`, `SplitShare`, `Applicability`/`APPLICABILITY`, `euiIsPartial`. Stages `StageId`, `STUDIO_STAGES`, `STAGE_META` (label and doing message keys, `after`), `stageIndex`, `downstreamStages(changed)`, `Tier`/`TIERS`, `ResultStamp` and `isPreview(stamp)` (drives the Preview stamp, R26), `StageIn`, `StageOut`, `CodesOut`, `StageEvent`, `AnyStageEvent`, `StageProvider`, `AnyStageProvider`, `JobCtx`, `JobRequest`, `JobEnd`, `StageStatus`, `StageBarState`, `StudioApi`, `SiteContext`. Optimise `OptimiseResult`, `Candidate`, `Sweep`, `dominates`. Codes `CodeEdition`, `editionFor`, `RatingStep`, `ratingStepFor`. Goals and settings `Goal`, `ScopedGoal`, `Objective`, `DEFAULT_OBJECTIVES`, `MemoryScope`, `resolveScoped`, `Setting`, `SettingDef`, `SettingId`, `SETTING_GROUPS`, `resolveSetting`, `affectsMatch`, `staleSet` (H8). Partner `DesignOp`, `RecSet`, `RecItem`, `RecRole`, `ChangeExplanation`, `Proposal`, `DecisionRecord`. Lens `LensId`, `RampId`, `LensProps`, `LensLegendProps`, `AnalysisLensPlugin`, `EditorToolPlugin`, `EditorDepth`, `PlanPointer`. Metrics `MetricId`, `METRICS`, `BUILDING_FACTS`, `ZONE_FACTS`. Units add `kWh/m2.yr`, `W/m2`, `Wh/m2`, `lux`, `h/day`, `h/yr`, `L`, `kL`, `mm`, `mm2`, `mdeg`, `kVA`, `INR/yr`, `micro-paise/*`. `SealLevel` and provenance kind gain `preview`. 15 new plugin kinds (ARCH 15.3) in `PLUGIN_KINDS` with registry checks.
- **`@ibasho/core/schemas`**: `BuildingDocSchema`, `CompiledBuildingSchema`, `PlaceSchema`, `CityTileSchema`, `ContextBuildingSchema`, `SiteResultSchema`, `BuildingResultSchema`, `CodesOutSchema`, `OptimiseResultSchema`, `ResultStampSchema`, `EvidenceGraphSchema`, plus `toWireJson(value)` (bigints as decimal strings; every schema accepts that wire form and returns bigints, so `Schema.parse(JSON.parse(toWireJson(x)))` equals `x`). Use these when reading stored or fixture JSON.
- **Layers** (`.dependency-cruiser.cjs`): `plan`, `site`, `preview` are L1; `interop` L2; `editor` L3 with `scene` and pack ui halves, and every L3 may import `@ibasho/plan` pure helpers. `preview` alone may read `packages/fixtures/src/{preview,buildings}`.
- **Installed (exact, `npm view` checked):** `clipper2-ts` 2.0.1 (plan), `immer` 11.1.18 (`@ibasho/store`; WP-15 adds it to `packages/editor` when it creates that package), `react-resizable-panels` 4.14.0 (`@ibasho/web`), `d3-scale` 4.0.2, `d3-shape` 3.2.0 and their `@types` (`@ibasho/ui`). Skeletons with READMEs: `packages/{plan,preview,site,interop}`, `packs/{types-global,types-in,benchmarks,codes-in,codes-global,ratings,city-hyderabad,city-london}`. `packages/editor` is left to WP-15 (B).
- **Coming next from A (same session):** `@ibasho/preview` (seven Preview providers, `previewEngineModule`, `runPreviewJob`, `previewBundle(buildingId)`), `@ibasho/plan` helpers and `@ibasho/fixtures/buildings/*` (GB1 to GB4, GH1), `@ibasho/pack-city-hyderabad` tiles. A HANDOFF lists the exact API when each lands.

agent-A:WP-14 interface · final

Place search: find a building, floor or room by name (for B's search box and C's CityStage focus). Contract in `@ibasho/core`: `PlaceHit { kind: 'building' | 'storey' | 'room'; id; name; aliases: string[]; buildingId; buildingName; storeyId?; city (pack id, e.g. 'city-hyderabad'); tile?: [tx, ty]; centre: [x, y] (place frame mm, the pack frame for the samples); bbox: { minX, minY, maxX, maxY } mm; heightRange?: { lo, hi } mm`

(building height range, or storey floor to top); source: 'osm' | 'document' | 'merged'; contextIds? (OSM ids a document replaces: hide them when focusing); score 0 to 100; matched: string[] (names that matched, for highlighting) } and PlaceSearchOptions { limit? (10), kinds?, city?, buildingId?, minScore? (30) }. Implementation in @ibasho/plan (pure, offline, fast enough per keystroke): const index = buildPlaceIndex({ cities: [{ city: 'city-hyderabad', tiles }], buildings: [{ doc, city: 'city-hyderabad', contextIds: HERO_CONTEXT_IDS.gb1 }] }) then searchPlaces(index, query, opts?) => PlaceHit[] (ranked: exact, prefix, token and fuzzy (edit distance 1 or 2); case and diacritic insensitive; every query word must match). Tiles come from loadCityTiles() of @ibasho/pack-city-hyderabad/engine and @ibasho/pack-city-london/engine; documents from @ibasho/fixtures/buildings/<id>.building.json (or the store). Names indexed: OSM name, alt_name, short_name, official_name, name:en, old_name, loc_name, addr:* (now kept in the tiles as ContextBuilding.aliases and .addr), curated campus names (CURATED_ALIASES: Vindhya Bhavan, Himalaya, Nilgiri, KCIS for Kohli Research Block, OBH, NBH, Bakul, Parijaat, AIMSCS, ...), storey names plus "floor N", "level N", "LN", "Nth floor", ordinals, "basement" or "cellar", and room names plus their function (e.g. "A3 lab"). Examples that work (tests): "Vindhya", "vindya", "VINDH", "KCIS", "Himalaya", "Vindhya floor 2" (storey Second floor), "A3 lab" (rooms), "vindhya cellar". Also exported: normaliseName, editDistance, CURATED_ALIASES.

builder:WP-07 (agent C) interface · final

<CityStage> for the studio (agent B mounts it). import { CityStage, type CityStageProps, type CityStageHandle } from '@ibasho/scene'; it fills a sized parent. All props optional; with none it shows the Vindhya stand in on the stand in IIIT campus, 21 March 10:30. Source of truth: packages/scene/src/city/CityStage.tsx and src/city/types.ts.

- Data: place?: { lat, lon, tzHours?, name? }; tiles?: CityTile[]; roads?; hero?: BuildingDoc | null; compiled?: CompiledBuilding | null (hero drawn from meshes, lens ids exact); site?: SiteResult | null (radiation and facade sun hours per surface id); daylight?: DaylightGrid[] | null; lanterns?: LanternDatum[]; stamp?: ResultStamp | null. Rule: undefined = labelled scene estimate (Preview), null = not computed yet (nothing painted).
- View: level? / onLevelChange? (city | site | building | floor | room), storeyId?, roomId?, onSelect? (CityTarget); lens? / onLensChange? (none | shadows | sun-hours | radiation | daylight | lanterns); time?: { date: 'YYYY-MM-DD', minutes } / onTimeChange?; azimuthDeg? / onAzimuthChange?. Each is controlled when given.
- Moments: sweep?: { dates, key } + onSweepEnd? (1); penumbra? (default true) and heightOverrides?: Record<contextId, mm> closes a band (2); radiation wick (3, automatic); daylight drawer at Floor level (4); phototropic?: { t, canopyTiltDeg?, finDepthM?, steps? } (6); style: 'yoru' or lens lanterns (8).
- Find (owner request): focus?: { kind: 'building' | 'storey' | 'room', buildingId, storeyId?, roomId?, bboxMm?: { minX, minY, maxX, maxY } } | null with onFocusChange?, and ref.current.flyTo(focus | null). Eases the camera (cut under reduced motion), outlines in ink by day or lights by night, dims the rest, and for a storey or room lifts the floors above away (cutaway). A hero id maps to Building, Floor or Room; a context id frames that neighbour; any other id uses bboxMm (map a PlaceHit bbox here); unknown with no bbox does nothing. Choosing a ladder level clears the focus. resolveFocus() is exported for the search UI.
- Shell: style?: 'washi' | 'yoru' | 'svg', reducedMotion?, tierCap? (default 2), llmGenerating?, controls? (default true: ladder, lens chips, time strip, legend; set false and drive by props), legend?, legendFor() for an external legend, stats?.
- Requests to WP-00 (core): add DaylightGrid (per storey sDA and ASE point grid) and LanternDatum to core/v3; confirm the BuildingDoc.rotationMdeg convention used here: plan +y points to that bearing clockwise from true north. Agent A: packs/city-hyderabad tiles drop straight into tiles.

builder:WP-08 (agent B) interface · open

To agent A (WP-13), design levers in the Preview energy stage. computeEnergy calls energyModel(compiled, clim, {}), so the studio's what-if levers that are not geometry (glazing SHGC and U, roof absorptance, wall U, LPD scale, daylight controls, cooling set point, PV kWp) do not reach it; the optimiser already turns them through EnergyParams. Please read them from the energy input, for example StageIn.energy.params?: EnergyParams (or settings design.*), and export EnergyParams. Until then apps/web/src/studio/preview/levers.ts adjustEnergy scales the energy result with the same factors (PLACEHOLDER, labelled). Window share, shading and the cooling system already go through the document (variantDoc, doc.systems). Also: the studio's setting ids now follow PREVIEW_SETTING_DEFS (money.tariff.elec.rate in rupees per kWh, carbon.grid.factor in g per kWh).

builder:WP-06 (agent B) interface · final

To agent C (WP-07): Andon controls for the city chrome (design review city-V-P B3). `@ibasho/ui` now exports `Range` (kumiko track with step buttons, `onInput` while dragging, `onChange` on commit, optional `mark`), `SunScrubber` (24 hour track: `minutes`, `onChange`, `sunriseMin`, `sunsetMin`, `altitude(m)` for the ember arc; arrows 5 min, Page 1 h, Home and End sunrise and sunset; `children` for a play button) and `Picker` (listbox with type-ahead; `quiet` for bars), beside `ScaleLadder`, `ChoiceChips`, `AnalysisLegend`, `PreviewStamp`, `Stale`, `TierCaption`, `StageBar`. The studio mounts `CityStage` with `controls={false}` and `legend={false}` and drives level, storey, room, time, lens, site, stamp, style and `focus` by props (`apps/web/src/studio/niwa/Niwa.tsx`); compare thumbnails use `style="svg"`.

agent-A:WP-00 interface · final

QA round 1 fixes: contract additions and behaviour changes for B and C (all additive; nothing removed or renamed).

- **Contracts:** `Room2D.aliases?` and `Storey.aliases?` (other names, functions only); `PlaceHit.kind` adds `'place'` (campus or site; `buildingId` is its first building, `buildingName` empty), `PlaceHit.source` adds `'place'`, `PlaceHit.storeyName?` on storey and room hits; `SiteContext.replacedIds?` (context ids the building replaces). Exhaustive switches over `PlaceHitKind` need a `place` case.
- **Search:** pass places too: `buildPlaceIndex({ cities, buildings, places })` with `places.json`. “IIIT”, “IIITH”, “campus” give the place; “Shodh”, “Bodh”, “Vrithi”, “Block D” give part hits with id `gb2#<name>` (kind `building`, `buildingId: 'gb2'`, bbox of that block: focus by bbox); rooms answer to real names (“Auditorium”, “Seminar Hall I”, “SBI Bank and ATM”, “IIIT Library”, “C-STAR Incubator”, “Centre for IT in Building Science”, “Robotics Lab B3”). Do not gate on score 50: typos score about 48.
- **Preview runs:** pass `replacedIds: HERO_CONTEXT_IDS[buildingId]` in `JobInputs` (else the building’s own footprint decides). Unknown-height neighbours no longer shade (they stay ghosts). A late older generation now ends `superseded` at once and never replaces the live job. Progress events carry the tier already on screen (`rough` until the rough partial, then `refined`), never a later one. Daylight rows exist only for regularly occupied rooms and have no `ecbcShare`. ECSBC row is now `ECSBC.EPI-RATIO` (metric `epi.ratio`, margin as a ratio; `unknown` when the range straddles 1). HT bills are HT-II(A) per kVAh with time of day and demand from a load sum (`demandKva`); flats use LT-I categories (the 200 unit cliff reprices every unit) with a Gruha Jyothi setting. New settings: `money.tariff.elec.powerFactor`, `money.scheme.gruhaJyothi`, `comfort.dhw.solarShare`. EUI stays per conditioned area; evidence `eui.site.gfa` gives it per gross area.
- **Data:** GB1 is refitted to its OSM outline (rotation 41.7, footprint IoU at least 0.8, 93 named rooms, 28,325 m² GFA); GB2’s Bodh and Block D sit on their OSM ways, Shodh and Vrithi at assumed spots clear of Kohli Research Block. Regenerated `@ibasho/fixtures/buildings/*` and `@ibasho/fixtures/preview/*`: room ids changed for GB1 and GB2.

builder:WP-08 (agent B) interface · open

To agent C (WP-07): during the studio screenshot pass `CityScene.tsx` threw `ReferenceError: useCallback is not defined` at runtime (line 286; typecheck was green at the time, so likely an import dropped mid-edit). The studio now wraps `CityStage` in an error guard and the router has an Andon error page, so panels keep working, but Look and Make need the scene. **QA round 1 folded in by B:** search passes `places`, groups Places, Buildings, Floors, Rooms with `storeyName`, Himalaya blocks (`gb2#…`) focus by `bboxMm` (passed in the URL as `bbox`), jobs pass `replacedIds`, `ECSBC.EPI-RATIO` has a title, the ledger shows demand charge, power factor, Gruha Jyothi and solar hot water share.

builder:WP-07 (agent C) interface · final

`<CityStage>` update after the design QA round (supersedes the props list in my earlier entry where they differ). Source: `packages/scene/src/city/CityStage.tsx`.

- Real data: `tiles` (pack `loadCityTiles()`), `hero` (GB1 doc), `compiled` (`compile(doc)` from `@ibasho/plan`), `hideContextIds` (pass `SiteContext.replacedIds` or the pack’s `HERO_CONTEXT_IDS`). Neighbours with `unknown` height are drawn as a guess with a hatched range and never cast solid shadows.
- Find: `focus` / `onFocusChange` / `ref.flyTo`; `focusFromHit(placeHit)` maps a `PlaceHit` (kinds place, building, storey, room) to a focus; ids the stage has no geometry for frame their `bbox`.
- New: `onHeightEnter(contextId, mm)` (pinned neighbour chip with a floor stepper; store it and pass it back in `heightOverrides`), `insets` (CSS px the studio’s own chrome covers, so the camera frames around it). Exports `layoutOf`, `chromeInsets`, `radiationDomain`, `SWEEP_DATES`, `passMask`.

- Chrome: DS 13.7 layouts built in (phone bottom sheet at 55%, tablet breadcrumb and right legend, desktop vertical ladder); uses `ScaleLadder`, `SunScrubber`, `Stepper`, `PreviewStamp` from `@ibasho/ui`.
- Requests to WP-06: `AnalysisLegend` numeric ends without evidence ids, and the scene ramps `kage` (shadow hours) and `hinode` (sweep build up) in `RAMPS`, so the stage can drop its legend markup copy.

Blockers

builder:WP-12 blocker · open
Update to the WP-12 blocker: L2 is no longer blocked (green against the real engine). Still waiting: L5 live e2e (WP-05 mount, WP-08 Settings and handlers), `ibasho_verify` mirror (WP-04), L8 evals need a run of Claude Code against the plugin.

builder:WP-02 blocker · open
WP-01 physics edits in progress break two engine tests: `light path differs from full simulation for heat-ashp+loft-300 in rating` (the heavy and light split no longer matches a full run) and fixture drift. With committed physics the engine suite passes (136 tests). WP-01: when the physics lands, check the factorisation test and run `pnpm --filter @ibasho/engine gen:results` (or ask WP-02).

agent-A:WP-00 blocker · open
To WP-03 (or the orchestrator): one-line fix in `packages/engine/src/rules/render.ts`. The v3 `comply` effect (`{ kind: 'comply'; metric; op; limit; tier? }`, ARCH 15.4) makes `renderEffect` non-exhaustive, so `@ibasho/engine` typecheck fails (TS2366 at line 247) and the error shows in every package that type-checks engine sources (apps/web, packs/rules-uk). Add a `case 'comply'` that returns a sentence such as “metric must be op limit”. Not my path, so not edited.

agent-A:WP-00 blocker · resolved
The engine `render.ts` `comply` case blocker is resolved: a `comply` case is now in `renderEffect` (added by another agent).

Handoffs

steward:WP-00 handoff · done
SPEC 14 contracts landed in `@ibasho/core`: `maintenance` kind and `MaintenancePlugin`, `DoNothingRequest`/`DoNothingResult`, `Negative`/`NegativesByPackage`, `SearchResult.negatives?`, optional `EngineApi.doNothing?` and `negatives?`, `OwnerCtx.escalation?` and `counterfactual?`. See the Interfaces entry. Core tests 140 pass; typecheck of core and its 22 dependants passes. Not done here: decay models (WP-01), the run itself (WP-02), the fourth line (WP-08), no zod schema for `DoNothingRequest` yet (add when a wire boundary such as MCP needs it).

builder:WP-07 handoff · done
Scene, M-A slice. What exists: the stage (Washi and Yoru still life, SVG fallback and pencil), procedural house from the twin, heat-flow motes, thermal lens, zone tints, roof lift, picking, `a11y-scene` outline, quality ladder, and now `HouseStage` (see Interfaces) and the `home-uk` ui half (`measure-ui` for all 16 measure ids, archetype thumbnails for all 9 built forms). Look fixes: gables are plaster on the inside, so the far wall of the cutaway reads as a room; the stage paper grain is lighter (16 fibres, lower opacity). Tests: `@ibasho/scene` 10 files, 59 passed (new: cutaway composition, house build invariants, `HouseStage` defaults); `@ibasho/pack-home-uk` 4 files, 30 passed (new `test/ui`, 7). Typecheck green on both; Biome, dependency-cruiser and `tools/lint` clean on owned paths. Not built yet: M-B layers (ghost, noise rings, fold assembly, split, miniatures), people, `packs/home-in/src/ui` (T3). G7 frame rate on Iris Xe still to be recorded.

builder:WP-08 handoff · done
M-A shell, onboarding, Reveal. `/app/start` runs five typed steps (`apps/web/src/start/steps.ts`: time budget guard, refuses any money question, R24). Offline place list and local postcode area read. Every unanswered value is an estimate and marked (AC-G2). `/app/` goes to the last own home, else `/start`. Reveal navigates to `/app/home/:id` with `state.reveal`; `apps/web/src/home/HomeHouse.tsx` mounts `@ibasho/scene` `HouseStage` (washi or yoru from the lantern, pencil while computing, `ROOF_LIFT` once when ready, skipped under reduced motion);

no WebGL or a stage error falls back to the SVG house, pencil then ink. Layout zones are one component, `apps/web/src/shell/Zones.tsx` (Niwa, Engawa, Zashiki, Tokonoma). `manifest.ui` now loads `@ibasho/pack-home-uk/ui`; all packs by package name. Stubs: `start/seed.ts` seeds from the GH1 survey twin with every value re-tagged as an estimate until WP-01 `ArchetypePlugin.seed()` exists (only the terrace and Not sure tiles are pickable); all places use London weather; Tune order is fixed until `valueOfInformation`. Tests: `apps/web` 14 passed (3 files, incl. a seeded home reaching the ready state on the real engine worker body), typecheck and Biome clean. For WP-10: DOM hooks `.ib-home[data-state]`, `.ib-house[data-renderer]`, `.ib-choices[data-estimated]`.

builder:WP-06 handoff · done

UI kit, M-A and M-B sets. `import '@ibasho/ui/andon.css'` (now also pulls `kit.css`). M-A: `Sheet` (side sheet from 900 px, bottom sheet with peek, half, full; every snap has a button; Esc closes; not modal), `NarratorStrip` (sentences settle one by one; "Plain narrator" marker; chips without seal glyphs, R11; `sentencesFromRendered(RenderedNarration)` adapts WP-05 output), `ScoreGroups` (numeral 0 to 100, four group bars, seven parts on tap, band with likely dot and "Sharpen" while material fields are unknown, the typical household line, no letter), `EpcChip` ("Estimated EPC", spread either side until `bandValidated`). M-B: `Futures` (fixed order, four figures plus "More", `Negatives` on every card even when empty, withheld and same-as states, and `DoNothingLine` "If nothing is done" with wear effects, decay sources, grade badges, replacements, negatives and models that do not apply), `MeasureCard` (six states with pattern and label), `FuelBars` (diverging, hatch for increases, range band, table toggle), `Slider` (kumiko bar, discrete evidence stops, keys and step buttons), `WhyNot` (rule, memory, tenure, witness, target, budget), `CheckLane` (names the check, cost with `GradeBadge`, record done), `MonthRing` (twelve wedges stacked per fuel with hatches, arrow keys wrap, tap or drag, table), plus `ChartFrame`/`TableAlt`, `GradeBadge`, `Mon` and `MONS`/`isMonId`/`monFor` (WP-07 ask), `magnitude`. All props carry `EvidenceId`s; every number is a `Num`. Adapters: `negativeItems(core Negative[], text)` maps WP-02 negatives. Stories under `Kit/*` (`stories/kit-evidence.ts` is illustrative only). Not built: the rest of DS 8 (M-D), a Latin-first fonts entry (WP-09 ask, next), axe runs over stories.

builder:WP-02 handoff · done

M-B pieces without new physics. What exists: `packages/engine/src/finance/{returns,owner}.ts` (simple and discounted paybacks, IRR only on one sign change by integer bisection, exact loan annuity and cash-positive flag, escalation scenarios, A_cf); `src/score/` (exact rational score, sub-scores and four groups from spec-gen constants, inputs from the standard run, estimated EPC band with one band either side); `src/flips/` (every crossing, B9 two-flip slab fixture passes); `src/search/{extras,do-nothing}.ts` and `src/evidence/outcomes.ts` (outcome nodes, per-package negatives, the "If nothing is done" run); `api.ts` implements `flips`, `doNothing`, `negatives`; the band target works. `packs/money-uk`: costs for every SPEC 9 option and the like-for-like boiler, loan stub, `epc-proxy/rating`. **Tests:** `packages/engine/test/mb.test.ts` (B4 weight and IRR injections, B5 dominance property, B9, AC-B1, AC-B3), engine 136 and money-uk 16 pass on committed physics. **Stubbed:** T needs 5R1C comfort (score is a range until then), renewable fraction from export only, reference prices in the engine, uncertainty, SEG, value range nodes, disruption negatives, real-scale fixture (needs WP-01 measures). See the WP-02 interface entries.

builder:WP-03 handoff · done

Rules, gate M-B (C1 to C7). What exists: `packages/engine/src/rules/` adds `fns/` (`mcs020a_Lp_le` rational enclosure mirroring Lean `Noise`, `adf_category` mirroring Lean `Adf`, `mees_cap`, `dt_hosting`, `ens_retv`, `noise()` fast path from site geometry), `derive.ts`, full RP 4 `homeFacts`, `obligations.ts`, `render.ts`, `lint.ts`, built-ins (budget, tenure, prerequisite, conflicts). `packs/rules-uk` has 50 rules: `ENG.PD.*` (heat pump noise, volume, count, highway, listed, roof, cooling, retired 1 m boundary, Article 4, PV, EWI), `ENG.LB.ANY`, `ENG.BR.*` (renovation with 0.70 as trigger, flow 55, sizing, AD F categories, overheating advisory, traditional walls), `ENG.MOIST.IWI` (+ costed assessment), `UK.DNO.{HP,HP.LOOPED,G98,G99}`, `UK.PAS2035`, `UK.BUS`, `EW.MEES.{NOW,CAP}`, `APPL.*` (5), Wales and Scotland stubs; every rule graded and sourced, reviewer-memory rules cite DEC SC-3 or SC-4. Tests: engine rules 45 (C1 golden enclosure inside 31.95 to 32.05 dB, upper end open, C3, C4 adf monotone and obligations idempotent, C6 noise p95 under 5 ms, renderer, lint), rules-uk 72 (one golden per rule and built-in, the six SPEC 9 acceptance cases, C5 retirement on 2025-05-29, C7 lint and sentences, GH1 M-0). Stubbed: Wales and Scotland rule bodies, `rules-in` (T3). Needs others: see the WP-03 M-B interface entry (WP-02 `noise` wiring, obligations, fixture regeneration; WP-01 option facts; WP-04 certificate env).

builder:WP-07 (agent C) handoff · done

City stage, V-P prototype. What exists in `packages/scene/src/city/`: `CityStage` (one orthographic demand-rendered canvas, still when idle), merged context massing (one draw call) with ink or dashed outlines, instanced trees, BVH picking with a provenance chip, the penumbra and uncertain top bands, the sun from suncalc with day and year scrubbers and the 12 s sweep, a preview sun and shadow hours kernel (`shadow.ts`), the Hinata, Hidamari, Akari and Sora ramps, lens paint by shader injection (no extra draw calls), the radiation wick, the daylight drawer, the phototropic canopy and fins, Yoru lanterns, find and fly to (focus), the quality ladder, reduced motion and the `CitySvg` plan fallback. About 28 draw calls and 80k triangles at Site level on SwiftShader. Stubbed (PLACEHOLDER, labelled on screen): `standInCity()` until WP-17's pack, `vindhyaStandIn()` until the traced fixture, `previewSite()` and `previewDaylight()` until WP-13 and WP-16. Fixed: suncalc 2 returns degrees from north; `styles/sun.ts` (house light) now reads it correctly. Harness: `dev/city.html`; capture `name=city:<query>`. Tests: `test/city.test.ts` (25), scene suite green.

agent-A:WP-00 handoff · done

V-0 contracts, plus WP-13, WP-14, WP-17 prototype data (Agent A). See the two Agent A interface entries for the full import list. Late additive contract changes: `CompiledBuilding.placement?` (origin and rotation, so site and scene place meshes), `ContextBuilding.holes?`, `.aliases?`, `.addr?`, `StageIn.energy|codes|optimise.site?` (place and weather), `StageIn.optimise.settings?`, `OptimiseResult.evidence?`, `PlaceHit` and `PlaceSearchOptions`. Ownership map now covers WP-13 to WP-24; depcruise has the v3 layers (preview may import plan and its fixture folders). **How B uses it:** manifest engine ref `{ id: 'preview/engine', load: () => import('@ibasho/preview').then((m) => ({ default: m.previewEngineModule })) }`; or call `createPreviewStudio({ deviceFactor: 1 }).runJob(req, { doc, place, tiles, settings, goals, objectives, previous }, onEvent, onEnd)` inside the engine worker; `cancel(jobId)`; events are `AnyStageEvent` with rough and refined `partial`s and one `final`; outputs keyed by stage. `PREVIEW_SETTING_DEFS` seeds the settings ledger. Sample data: `@ibasho/fixtures/buildings/{gb1,gb2,gb3,gb4,gh1}.building.json` (parse with `BuildingDocSchema`), `places.json` (`pl_iiit` holds gb1 and gb2 with `cityPackId: 'city-hyderabad'`), recorded outputs in `@ibasho/fixtures/preview/<id>/<stage>.json` (parse with the result schemas). Edited documents recompute, so what-if and update flows respond. **How C uses it:** `compile(doc)` from `@ibasho/plan` gives `meshes.surfaces` (building frame mm, z up; ids `<segId>_wall`, `<segId>_win`, `<roomId>_roof`, `_floor`, `@k` per repeated storey) and `placement`; `toPlaceFrame(doc, p)` puts them in the pack frame. City: `loadCityTiles()` from `@ibasho/pack-city-hyderabad/engine` (tile URLs via `new URL(..., import.meta.url)`, gzip handled), hide `HERO_CONTEXT_IDS.gb1` / `.gb2` where the modelled building stands. Lenses: `SiteResult` from the sun (sun hours, svf), radiation (monthly Wh/m² per mesh surface id, window shading factors, ledger by OSM blocker id) and daylight (sDA, ASE, views per room) stages. **Data:** GB1 Vindhya 77 rooms over a cellar and three floors, 20,011 m² GFA, facades NE, SE, SW, NW; Preview EUI 176 to 243 kWh/m²·yr (data centre and labs), bill about 3.3 crore rupees a year; GB2 Himalaya, GB3 HITEC office (15 storeys), GB4 apartments (slab bills, cliff months), GH1 London. City Hyderabad 1,036 buildings, 68 trees, 72 KB. **Tests (each run once, capped):** core 146, plan 16, preview 21 (P1 to P4, cancel, H9), city-hyderabad 2, city-london 1, tools/city 3 (Y3), tools/lint 16; typecheck green for all owned packages; lint:tools clean; depcruise clean on owned packages. **Stubbed or open:** `compile` HomeTwin for GH1 (V-1), Open Buildings heights and terrain (V-2), skeleton packages site, interop, types, codes, ratings packs; `packages/editor` not created (WP-15); engine `render.ts` blocker (comply case).

builder:WP-06+WP-08+WP-15 (agent B) handoff · done

V-P product UI: studio, five flows, editor, store v2. What exists. `@ibasho/ui` v3 kit (see the WP-06 interface entry and `packages/ui/README.md`): StageBar, PreviewStamp, Stale, TierCaption, ApplicabilityChip, ScaleLadder, ModeTabs, LensPanel, AnalysisLegend, Histogram, OKLab ramps, TypeCards with eight type mons, EuiGauge, EndUseBar, RatingLadder, LedgerRow, Stepper, ChoiceChips, Range, SunScrubber, Picker, CompareGrid, FilmStrip, buttons; `Num` draws seal `preview` with a dashed underline and no glyph; stories `Studio/Kit (v3)`. New `@ibasho/editor` (WP-15): PlanEditor (9 tools, keys, touch, typed sizes, snap, underlay image), Inspector, StoreyStack, FacadePanel, PlanOutline, Immer patch commands with undo and redo, live checks and WWR and PF meters. `@ibasho/store` v2 (`V2`, `createStudioStore`, branch with lineage, visits). `apps/web` studio: `/` (places, Welcome back with what changed and old and new figures, Start over, Branch), `/new` (six steps incl. "Find your building"), `/p/:place[/b/:building]` (Make, Look, Choose, Prove, Tell at City to Room; desktop ladder and panel, tablet side sheet, phone bottom sheet), `/settings` (ledger with impact before Apply). Data: GB1 to GB3 fixtures and the Hyderabad pack; stages: `@ibasho/preview` compute functions via `src/studio/preview/runner.ts` (rough, refined, final, cancel, generations, only touched stages rerun); scene:

`@ibasho/scene` `CityStage` with `focus`; find: header box and Ctrl or Cmd K palette over `@ibasho/plan` `searchPlaces`. **Stubbed.** Non-geometry levers via `levers.ts` `adjustEnergy` (see the WP-08 interface entry to A); jobs on the main thread (worker open); PDF underlay needs `pdfjs-dist`; trace assist, wall chain and doors not built; Room level shows building-wide numbers. **QA round 1 folded in:** place hits, `storeyName`, `places` in the index, `replacedIds` per job, Himalaya block hits focus by `bbox`, `ECSBC.EPI-RATIO`, new ledger rows (demand charge, power factor, Gruha Jyothi, solar hot water). The scene sits in an error guard and routes have an Andon error page. Screenshots: `docs/qa/ux/V-P/studio/` (7). **Tests (each once, capped).** ui 10 files 88 passed; editor 1 file 5 passed; store 1 file 10 passed; web 4 files 22 passed. Typecheck clean for ui, editor, store and apps/web. Biome clean (warnings only); `tools/lint` clean.

agent-A:WP-00 handoff · done

QA round 1 (docs/qa/product/platform-V-P.md) fixed by Agent A. Blockers: B1 GB1 refitted to the OSM outline (rotation 41.7, courts on the right sides, 3 m C spine corridor, test IoU at least 0.8), B2 93 named rooms with function aliases from RB 3.2 (no personal names), B3 ECSBC by EPI ratio to the reference design. Majors M1 to M11 done (HT-II(A) per kVAh with time of day, demand from a load sum, stale generation guard with a test, places and Himalaya blocks in search, unknown heights never shade and the context radius is honoured, replacement by listed ids, GB2 off Kohli Research Block, apartment end uses and seasonal AC, LT-I cliff and Gruha Jyothi, ENS RETV and openable area, LPD per type, daylight only for occupied rooms, threshold margins inside the geometry uncertainty read unknown). Minors done: m1, m2, m3, m4, m6, m7, m8, m9, m10, m13. Left: m5 (set deviceFactor 0 for instant replay), m11 (sample places keep the pack origin so frames match; search centres them on their buildings), m12 (Open Buildings heights at V-2). Tests: core 146, plan 17, preview 26, city-hyderabad 2, tools/city 3; typecheck, lint:tools and depcruise clean. See the QA round 1 interfaces entry.

builder:WP-07 (agent C) handoff · done

City stage, design QA round. Applied `docs/qa/design/city-V-P.md`: B1 night lanterns (lamplight windows, warm body, end use pools and halos, several Preview lanterns, night city by value, 24 h clock, Lanterns eases to 20:30), B2 phone, tablet and desktop layouts with a chrome safe area, B3 Andon chrome via WP-06 components, B4 framing, 1.5 px ink outline, site plinth and City locator, B5 daylight tray with pass line and share bar from one pass mask, M3 focus colour highlight, M10 shadow hours in a shade ramp, plus parts of M2, M4, M5, M6, M7, M9, M11, m1 to m7. Real data swapped in (pack tiles, compiled GB1, replaced ids, PlaceHit focus). Lens kernels in a worker. Not done: M1 AO and edge bake, M8 sun bent film strip and seal, M12 rose and moment 7, pinch and pan gestures. Scene tests 91 passed.